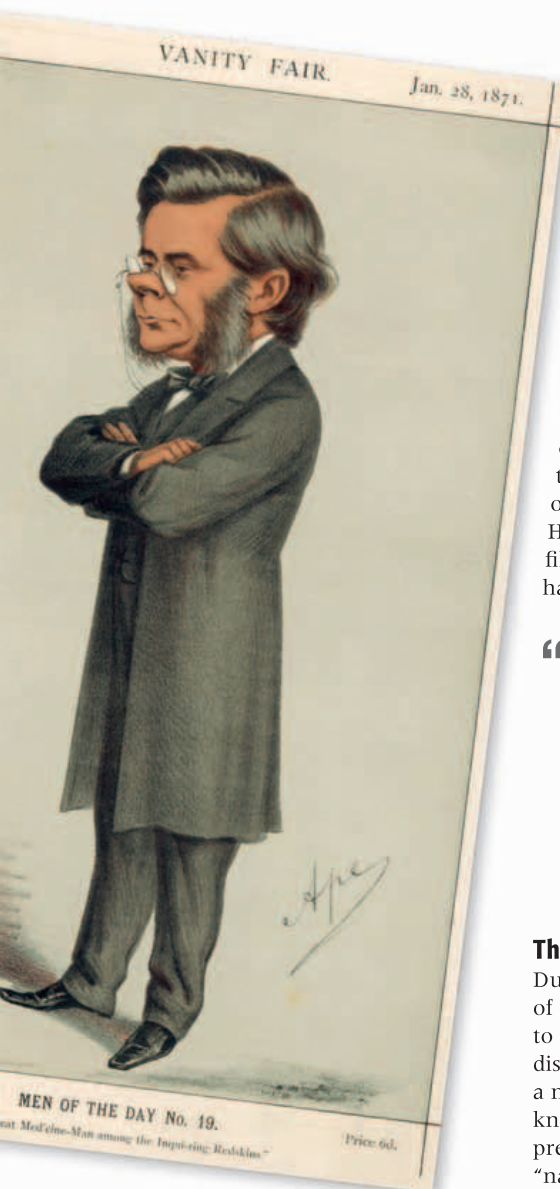




metaphorical, with the six days of creation being interpreted as epochs, or geological time periods. The idea that humans themselves were descended from apes—derided as the “monkey theory”—was much harder to accept. Some found a way to reconcile the information with their religious beliefs. The novelist and theologian Charles Kingsley wrote: “I have gradually learnt to see that it is just as noble a conception of the Deity to believe that He created primal forms capable of self-development as to believe that He required a fresh intervention to fill the [voids] which He himself had made.”

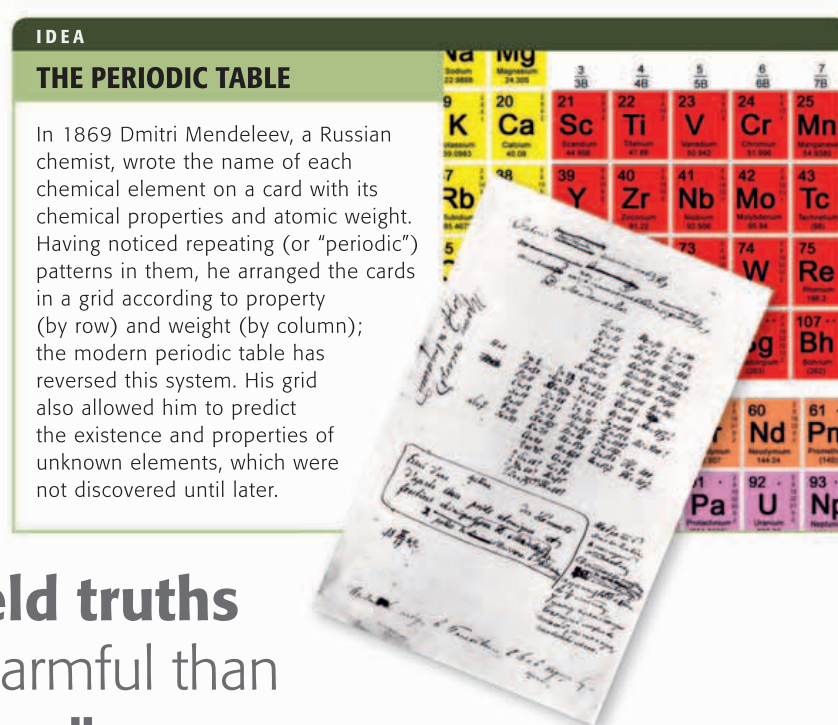
“Irrationally held truths may be more harmful than reasoned errors.”

T. H. HUXLEY, “THE COMING OF AGE OF ‘THE ORIGIN OF SPECIES’,” 1880

The march of science

During the 19th century, the elevation of science from a hobby of the eccentric to a respected and even admired discipline was marked by the use of a new word: its practitioners became known as “scientists,” where previously they had been called “natural philosophers.”

Science gained credibility alongside religion, rather than instead of it, and Christianity proved resilient to change. The reduction of the Church’s influence by the state following the French Revolution in 1789 (see pp.302–03) resulted in greater religious freedom throughout Europe, with a move away from state religions toward more numerous denominations. Competition between them resulted in the strengthening of religious loyalties, while a rise in atheism in urban areas



led to increased church building in many European cities.

Scientific progress continued, with people experiencing its effects through technology and medical advances. Governments began to address the need for scientific education: research-based university departments were established, as were scientific and technical institutions. As scientific knowledge filtered through society, it became more accessible. Despite the fears of the Church, society was not faced with a stark choice between science or God. Instead, these advances led in many cases to the acceptance of both science and belief.



Geological evidence

The discovery of fossilized ancient creatures such as these ammonites (prehistoric mollusks) seriously undermined the teachings of the Bible regarding the age of Earth and the way it was created.

rock and fossil evidence that suggested that Earth was much older than had been thought, casting doubt on Biblical chronology and the story of creation, though 19th-century geology was not sufficiently developed to make accurate estimates of Earth’s age.

The Church was able to cope with these new theories relatively easily. Many Christians were willing to consider Biblical chronology as

Science has continued its attempts to explain the universe, while the issue of Creationism still causes debate today.

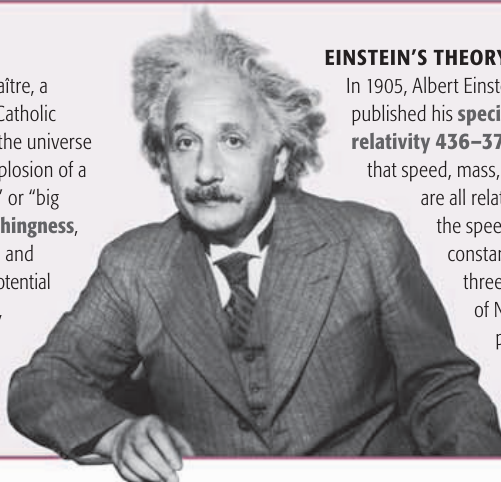
“Science without religion is **lame**, religion without science is **blind**.”

ALBERT EINSTEIN, PAPER FOR CONFERENCE ON SCIENCE, 1940

BIG BANG

In 1927 Georges Lemaitre, a Belgian scientist and Catholic priest, proposed that the universe was created by the explosion of a “quantum singularity,” or “big bang”—a state of nothingness, existing outside space and time, containing the potential for everything there is, has been, or will be.

ALBERT EINSTEIN



EINSTEIN’S THEORY

In 1905, Albert Einstein (left) published his **special theory of relativity** 436–37 >>. His theory that speed, mass, space, and time are all relative, and that only the speed of light is constant, replaced the three dimensions of Newtonian physics with a “space-time continuum.”

CREATIONISM The belief that the universe was created by a god, rather than being formed by a process of evolution.

CREATIONISM

In mainstream Western thinking, scientific theories have largely replaced Creationism. However, there is still a vociferous creationist minority, and in modern usage Creationism usually refers to Christian fundamentalists who oppose the teaching of scientific theories such as evolution and the Big Bang.

AFTER >>